

Intro to Programming

C/C++

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Learning Goals

- Comfort with
 - Basics of C/C++
 - Basics of Computer Science
- Familiarity with IDE
 - Visual Studio Community Edition
- Thinking logically
 - i.e. One step at a time
 - And Visualizing how computer works
- Independence of technology

Why C/C++

- Foundational understanding
 - Understand Computer Science
- Speed and control
 - Fastest programming language
- Really small programming language
 - C has 32 keywords
 - C++ has 92 keywords as of 2023
- Makes you digital native

When NOT C/C++?

- Slower manual speed of writing code
- Don't care about speed
- Don't care about deep Computer Science
 - Although this may not be achievable

Bit : Smallest unit of memory

- Bits are used for representing everything
- Have 2 states : 0 and 1 , like a bulb
 - On : 1
 - Off : 0
- Nibble : 4 bits
- Byte : 8 bits
- int (integer) 4 bytes
- char (character) 1 byte

Hertz : Unit of time and speed in Computers

- 1 Hertz : once per second
 - 1 unit of work per clock instruction
- Modern processors
 - Measured in Giga hertz
 - High Core Counts
 - More instructions

Hello World

- Let's go ,
- Program to print “Hello World”

```
#include <iostream>

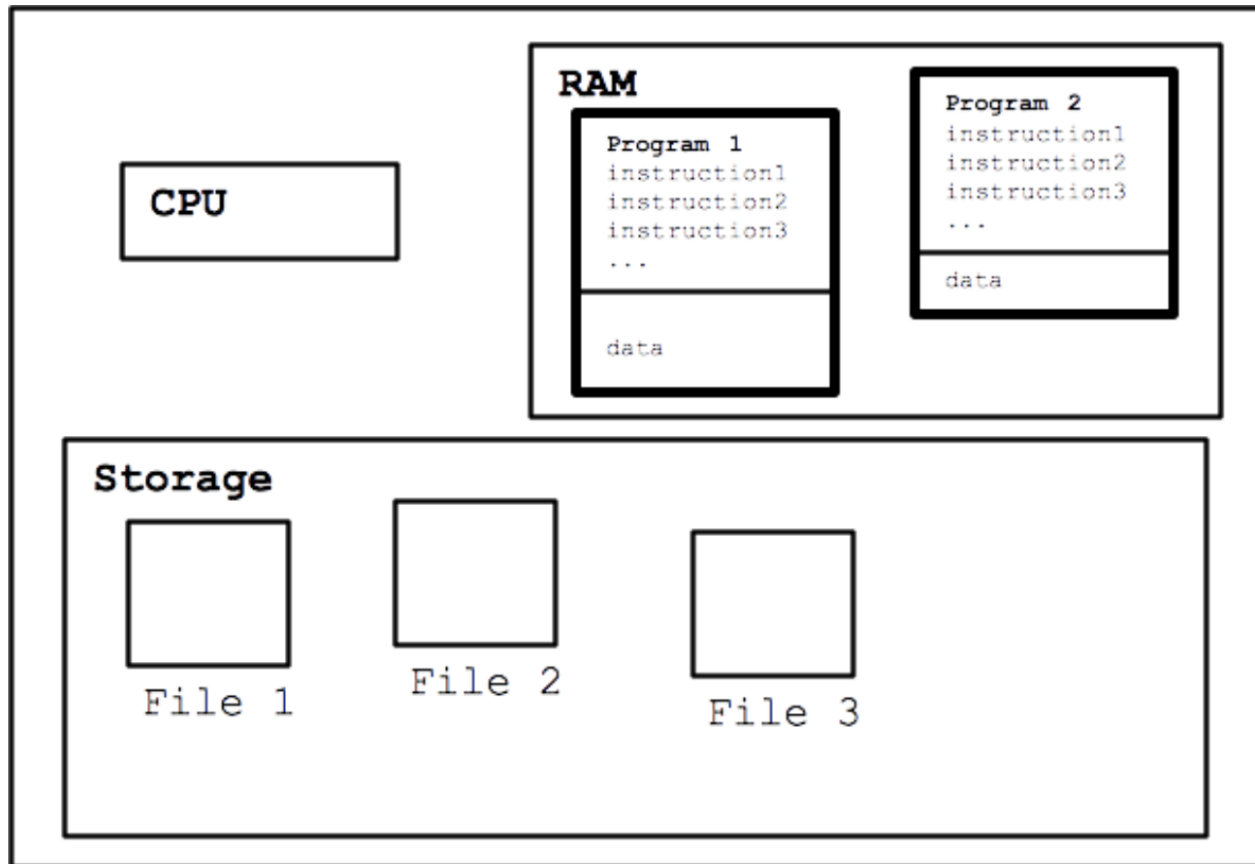
using namespace std;

int main()
{
    cout << “Hello World”;
    return 0;
}
```

Print a diamond pattern

```
  *  
 ***  
*****  
 ***  
  *
```


Running the program : model



Data Types_(Primitive/built in)

- bool
- int
 - short(2), long(4)
 - signed,unsigned
- float
 - float(4), double(8) , long double(8,10,16)
- char(1)
 - signed,unsigned
 - wchar_t(2)
- void

Variables

- Containers for storing data
- Value can change during execution (unless you don't want it to)
- Declaration
 - `int birthyear;`
 - `float weight;`
 - `char courseGrade;`

Data Types (Derived)

- Arrays
 - `char name[100];`
 - Size 100
 - Index : 0 - 99
 - `int age[10];`
 - Size 10
 - Index : 0 - 9
 - `float power[20]`
 - Size 20
 - Index : 0 - 19

Integer vs floating math

- Division
 - Float :continuous, contains decimal point
 - Int : discrete , truncates everything after decimal ,
- `float f = 10;`
 - `cout << f/3;`
 - 3.33
- `int i = 10;`
 - `cout << i/3;`
 - 3

For reference

no need to

memorize

Type Name

Bytes

Other Names

Range of Values

int	4	signed	-2,147,483,648 to 2,147,483,647
unsigned int	4	unsigned	0 to 4,294,967,295
bool	1	none	false or true
char	1	none	-128 to 127 by default
signed char	1	none	-128 to 127
unsigned char	1	none	0 to 255
short	2	short int, signed short int	-32,768 to 32,767
unsigned short	2	unsigned short int	0 to 65,535
long	4	long int, signed long int	-2,147,483,648 to 2,147,483,647
unsigned long	4	unsigned long int	0 to 4,294,967,295
long long	8	None	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
unsigned long long	8	none	0 to 18,446,744,073,709,551,615
enum	varies	None	
float	4	None	3.4E +/- 38 (seven digits)
double	8	None	1.7E +/- 308 (fifteen digits)
long double	8	None	Same as double
wchar_t	2		0 to 65,535

User Defined Data Type

- Enum
 - Short for enumeration
 - Define a set of named, integer constants

```
enum ESPEED  
{  
    low,  
    medium,  
    high  
};
```

```
ESPEED fanSpeed;  
fanSpeed = low;
```

Code : Greeting with name/age

- Given user
 - Store name in char array (`char name[100]`)
 - Store year of birth in int(`int birthyear`)
- Calculate age
 - `age = currentYear – birthyear;`
- Print Name, age.
 - `cout << "Hello "`
 `<< name`
 `<< " I know your age is :"`
 `<< age;`

Advanced : Count digits in a number

- Define a number
- Return the number of digits, (assume positive)
 - 129 $\rightarrow$ 3
 - 34 $\rightarrow$ 2
 - 0 $\rightarrow$ 1
- Hints
 - You need to know loops
 - while
 - You need to know integer maths
 - division

Input , Output and Processing for Humans

- Speech
 - Process → Speak → Listen
- Book
 - Read → Process → Memorize
- Conversation
 - Listen → Speak → Listen → Speak

I/O and processing for Computers

- Video games
 - Input (controllers) → Process → Output (Screen)
- Movies
 - Input (network) → Process → Output (visuals, audio)
- Console
 - Input(char, int, float) → Process → Output (char(s), int, float)

Standard Input

- cin
 - Read data from keyboard
 - Store it in variables
- Extraction operator
 - >>
- Can use multiple data types (char, int, float , ...)
- Example
 - int age;
 - cin >> age;

Standard Output

- `cout`
 - Write data to console/Screen
 - Reads from memory
- Insertion operator
 - `<<`
- Can use multiple data types(variables, literals, constants)
- Example
 - `int age = 172;`
 - `cout << age;`

Operators >>, <<

- Extraction and Insertion operators
- Can be cascaded
 - `cin >> age >> name;`
 - `cout << age << name;`
- << works with stream modifiers
 - “\n” : newline
 - `cout << “\n”; //moves the cursor to new line`
 - Or
`cout << endl; //Same visual effect as “\n” but is different`
 - (there are other stream modifiers too)

Lab

- Using Cin (Hint: define a variable first)
 - Input a character (char)
 - Input an integer(int)
 - Input a decimal(float)
- Using cout
 - Output a character
 - Output an integer
 - Output a decimal.
- Use endl and “\n”
- Cascade the operators

Assignment : Mad libs story

```
#include <iostream> // Required for input/output operations (cout, cin)
#include <string>    // Required for using string data type

int main() {
    std::string adjective1, noun1, verb1, adjective2, noun2;
    // Prompt the user for input and store it in the variables
    std::cout << "Enter an adjective: ";
    std::cin >> adjective1;

    std::cout << "Enter a noun: ";
    std::cin >> noun1;

    std::cout << "Enter a verb: ";
    std::cin >> verb1;

    std::cout << "Enter another adjective: ";
    std::cin >> adjective2;

    std::cout << "Enter another noun: ";
    std::cin >> noun2;

    // Construct and print the Mad Libs story
    std::cout << "\n--- Your Mad Libs Story ---\n";
    std::cout << "Once upon a time, there was a " << adjective1 << " " << noun1
        << ". It loved to " << verb1 << " all day long.\n"
        << "One day, it met a " << adjective2 << " " << noun2
        << ", and they lived happily ever after.\n";

    return 0; // Indicate successful program execution
}
```


Boss Assignment

- Input student details
 - Student Name
 - Subject name
 - Marks (out of 100)
- Process
 - Find grade using this table
 - $90 < \text{marks} \rightarrow A$
 - $75 \leq \text{marks} < 90 \rightarrow B$
 - $60 \leq \text{marks} \leq 74 \rightarrow C$
 - $\text{marks} < 60 \rightarrow D$
- Output
 - Grade for the student
- Challenge
 - Enter multiple students, print how many students had A, B , C and D grades each.
- Hint
 - Need to know conditional (if-else)
 - May need to know loop (while)

Recall , Review

- C/C++Keywords we know already
 - And few more
 - signed , unsigned
 - short , long
- Computers think in 0s and 1s
 - What types ?
- Computer have speed measured in hertz (Hz)
 - How much faster is 1Kilo Hertz than 1Hz ?
 - What are current computer speeds ?

Measuring computer capabilities (some more units)

- FLOPS : floating point operations per second
 - FP 16, **FP 32**, FP 64
- IOPS : Input/Output operations per second
- Fun facts
 - Computers are afraid of floats
 - Computers are afraid of division

Computer trends

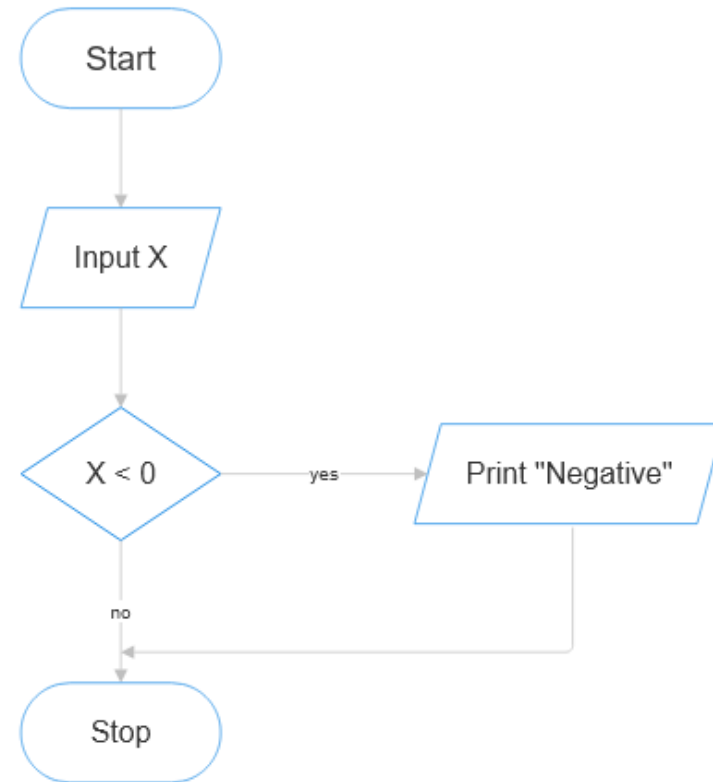
- Getting smaller
 - **Die** sizes have been shrinking
- Getting faster
 - Same size **die** have more **transistors**
- Getting crowded
 - More **core** counts per **die**
- Getting chatty
 - Networked , internet connected
- Getting efficient
 - More performance per unit of **power**

Back to C++, Decision Making & Branching

- if , else
- switch
- goto

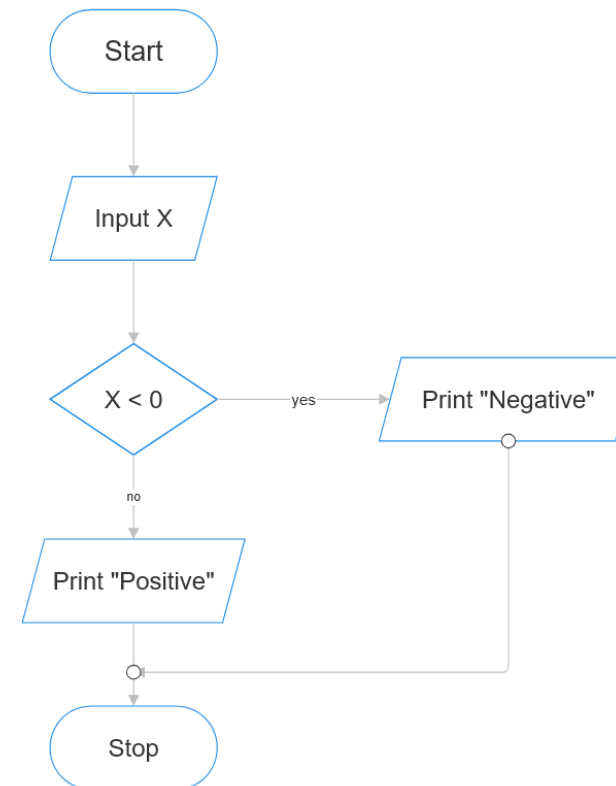
Making Decisions , using “if”

```
int x;  
cout << "Enter a number :";  
cin >> x;  
  
if (x < 0)  
{  
    cout << "\nThis is a negative number";  
}
```



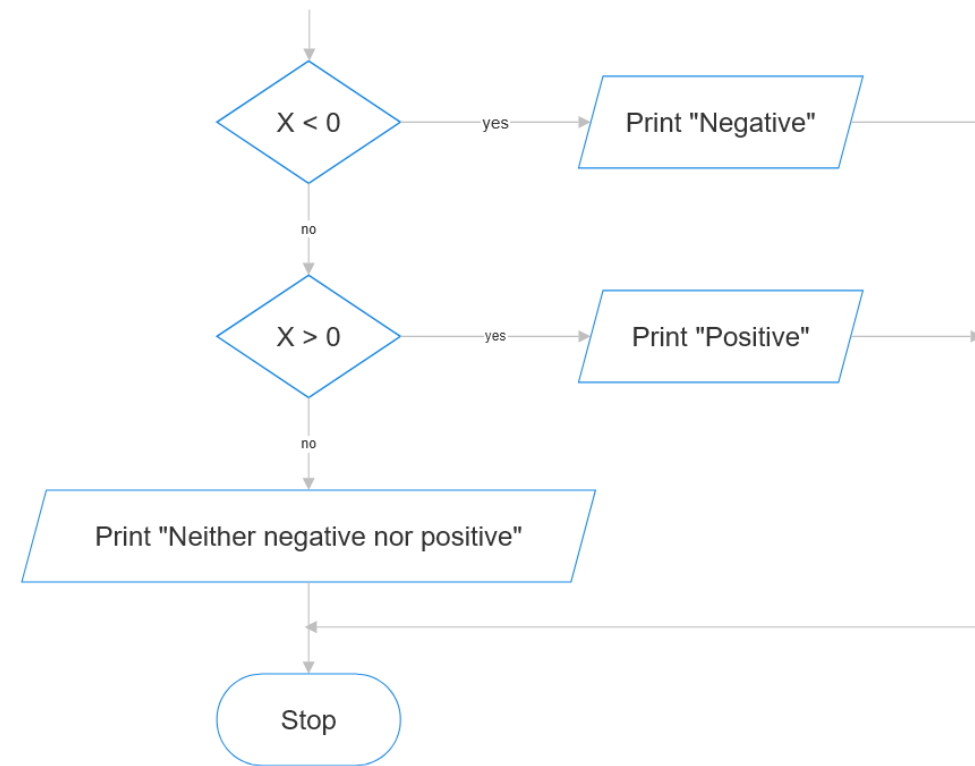
Making Decisions using “if else”

```
int x;  
cout << "Enter a number :";  
cin >> x;  
  
if (x < 0)  
{  
    cout << "\nThis is a negative number";  
}  
else  
{  
    //Bug ?  
    cout << "\nThis is a positive number";  
}
```



Making decision, If else if else

```
int x;  
cout << "Enter a number :";  
cin >> x;  
  
if (x < 0)  
{  
    cout << "\nThis is a negative number";  
}  
else if(x > 0)  
{  
    cout << "\nThis is a positive number";  
}  
else  
{  
    cout << "\nThis is a neither negative or positive";  
}
```

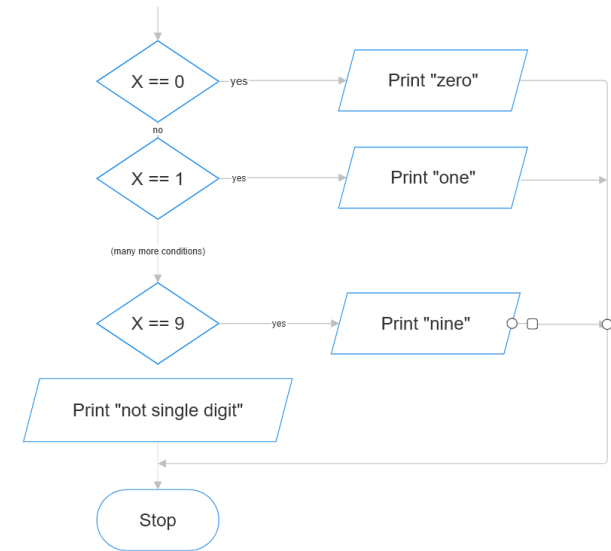


Relational operators

- less than
 - <
- greater than
 - >
- less then or equal to
 - <=
- greater than or equal to
 - >=
- is equal to
 - ==
- is not equal to
 - !=

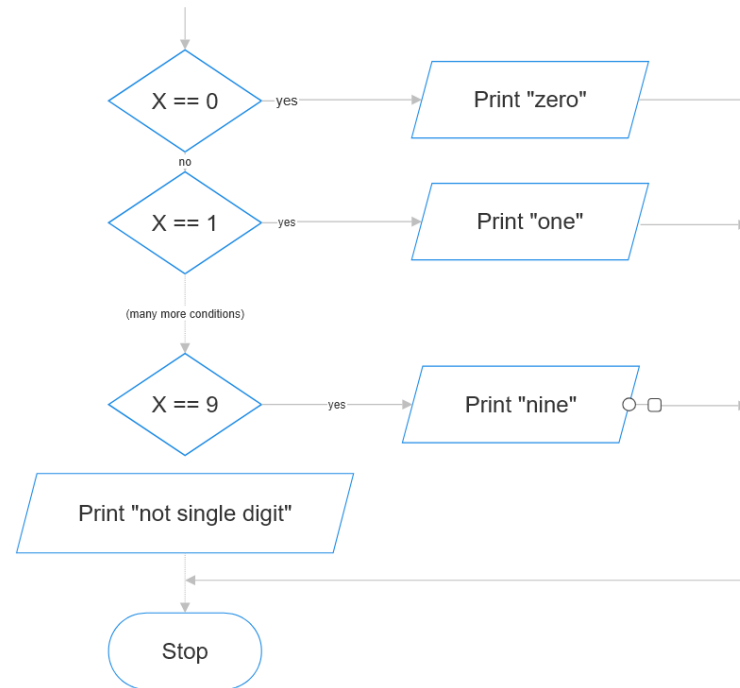
Lab (if else)

- Get user to enter a single digit number
- Print the digit in English
 - 0 → “zero”
 - 1 → “one”
 - ...
 - 8 → “eight”
 - 9 → “nine”
 - (Anything else) “→ this is not a single digit number”



Branching using Switch

```
switch (x)
{
    case 0:
        cout << "zero \n";
        break;
    case 1:
        cout << "one \n";
        break;
    case 2:
        cout << "two \n";
        break;
    case 3:
        cout << "three \n";
        break;
    case 4:
        cout << "four \n";
        break;
    case 5:
        cout << "five \n";
        break;
    case 6:
        cout << "six \n";
        break;
    case 7:
        cout << "seven \n";
        break;
    case 8:
        cout << "eight \n";
        break;
    case 9:
        cout << "nine \n";
        break;
    default:
        cout << "not a single digit number \n";
}
```



Lab (switch)

- Get user to enter a single digit number from [1,2,3,4,5,6,7]
- Convert it into day of week
 - 1 → “Sunday”
 - 2 → “Monday”
 - 3 → “Tuesday”
 - 4 → “Wednesday”
 - 5 → “Thursday”
 - 6 → “Friday”
 - 7 → “Saturday”
 - (Anything else) → “Invalid number for a day”

Asssignment , quiz game

- Create a quiz game containing atleast 3 questions.
 - Print a question ,
 - provide 4 numbered options ,
 - User enters the option number and verifies result.
 - Keep score of how many correct responses user input
- Print the score (e.g.3 /4 correct ,)
- An example
Q1. What is the capital of USA ?
 1. Seattle
 2. Los Angeles
 3. Washington DC
 4. ChicagoEnter your response : _

Boss Assignment

Convert a string of 0,1 to decimal

- Get a string of 0s and 1s from user
- Calculate what the decimal number for it is ?
- Note
 - This is very hard , may take hours/days/weeks
- Hint
 - You would need to know
 - string (or char array)
 - Loops (while)
 - Binary logic
 - *Loops within loops

Code, as seen by computer : HelloWorld

- Cpp vs assembly

```
1  #include <iostream>
2
3  int main()
4  {
5      std::cout << "Hello World!\n";
6  }
```

```
1 section .data
2     msg db 'Hello, World!', 0xA ; Message string with newline
3     len equ $ - msg           ; Length of the string
4 section .text
5     global _start
6 _start:
7     mov eax, 4                ; System call number for sys_write
8     mov ebx, 1                ; File descriptor (stdout)
9     mov ecx, msg              ; Pointer to the message
10    mov edx, len               ; Length of the message
11    int 0x80                   ; Call kernel
12    mov eax, 1                ; System call number for sys_exit
13    xor ebx, ebx               ; Exit code 0
14    int 0x80                   ; Call kernel
```

Code , as seen by computer : Find larger num.

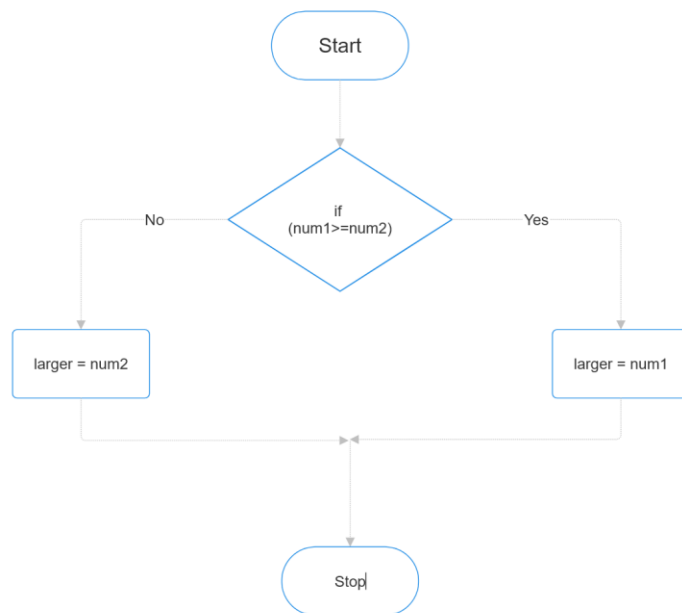
- Cpp vs Assembly

```
1  #include <iostream>
2
3  int main()
4  {
5      int num1 = 10;
6      int num2 = 20;
7      int larger = 0;
8
9      if (num1 >= num2)
10     {
11         larger = num1;
12     }
13     else
14     {
15         larger = num2;
16     }
17 }
```

```
1 section .data
2     num1 dd 10
3     num2 dd 20
4     larger dd 0
5 section .text
6     global _start
7 _start:
8     mov eax, [num1]    ; Load num1 into EAX
9     mov ebx, [num2]    ; Load num2 into EBX
10    cmp eax, ebx       ; Compare EAX and EBX
11    jge store_eax       ; If EAX >= EBX, jump to store_eax
12    mov eax, ebx        ; Otherwise, move EBX into EAX
13 store_eax:
14    mov [larger], eax   ; Store the larger value in memory
15    mov eax, 1          ; Exit system call
16    xor ebx, ebx        ; Exit code 0
17    int 0x80
```


Branching visualized

- Code looks sequential
- But really has branches



```
1  #include <iostream>
2
3  int main()
4  {
5      int num1 = 10;
6      int num2 = 20;
7      int larger = 0;
8
9      if (num1 >= num2)
10     {
11         larger = num1;
12     }
13     else
14     {
15         larger = num2;
16     }
17 }
```

```
1  section .data
2      num1 dd 10
3      num2 dd 20
4      larger dd 0
5  section .text
6      global _start
7  _start:
8      mov eax, [num1]    ; Load num1 into EAX
9      mov ebx, [num2]    ; Load num2 into EBX
10     cmp eax, ebx       ; Compare EAX and EBX
11     jge store_eax      ; If EAX >= EBX, jump to store_eax
12     mov eax, ebx       ; Otherwise, move EBX into EAX
13  store_eax:
14     mov [larger], eax   ; Store the larger value in memory
15     mov eax, 1          ; Exit system call
16     xor ebx, ebx        ; Exit code 0
17     int 0x80
```

Loops : Things that happen over and over ...

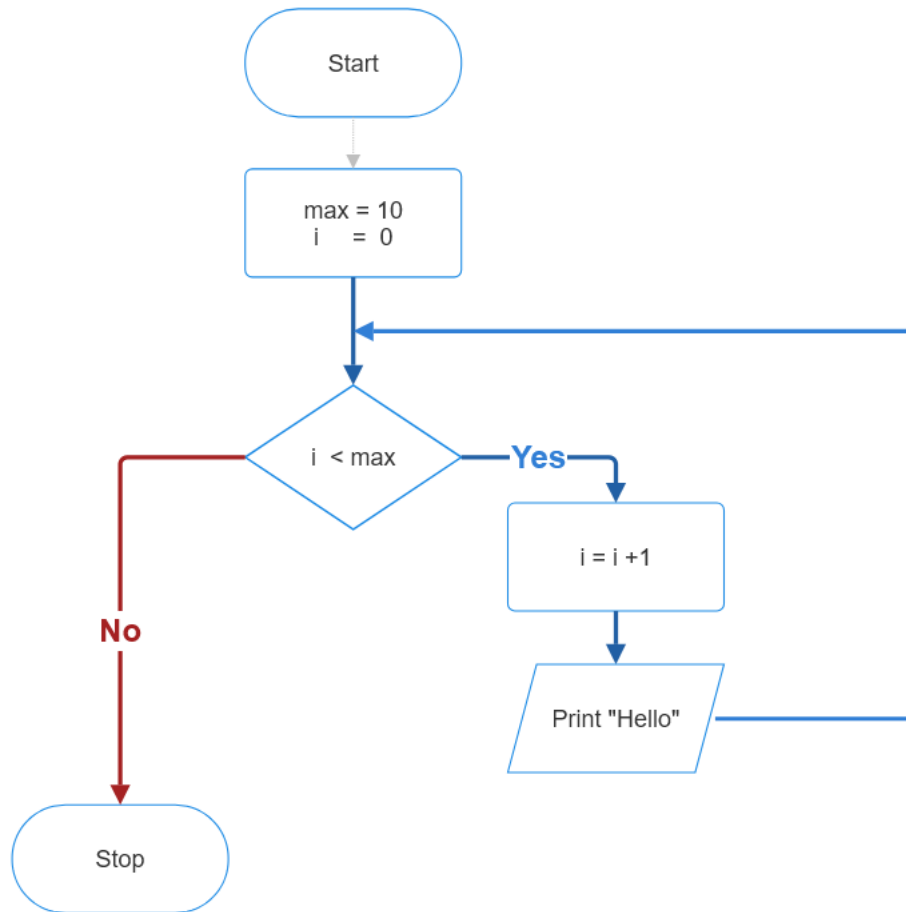
- Years
 - ...,2024,2025, ...
- Seasons
 - Spring, summer, fall, winter
- School
 - Learn , Homework, Tests
 - Scoring
- Reading a book
 - Title,... page, page,page ...,the end

Decision making and looping

- Loops
 - Print “Hello” 10 times
 - 10 x cout ?
- What if 100 times ?
 - 100 x cout ??
- Can we do better ?

Loop : visualized

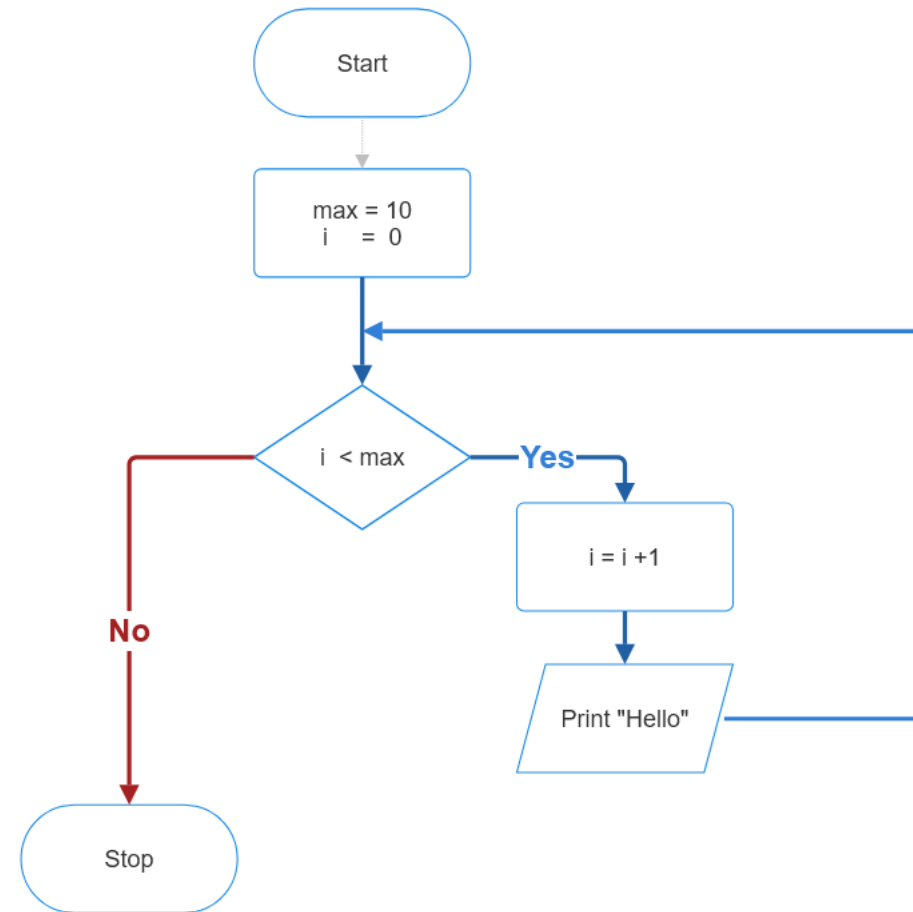
- Print “Hello” 10 times



Loop : visualized

- Print “Hello” 10 times

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int max = 10;
7      int i = 0;
8
9      while (i < max)
10     {
11         cout << "Hello " << endl;
12     }
13 }
```



Assignment 1, Guess the number

- Computer finds a random number between 0..9
- User guesses it
- If user fails .. they try again ... (*use while loop)

```
1  ✓
2  ✓ #include <iostream>
3  ✓ #include <cstdlib>
4  ✓ using namespace std;
5
6
7  ✓ int main()
8  {
9      int randomNumber = rand() % 10;
10
11     int num;
12
13     cin >> num;
14
15     //Fill in the rest ....
```

Boss Assignment

- Get a number from user
- Convert the number(int) into binary.
- Examples
 - 0 → 0
 - 1 → 1
 - 7 → 111
 - 8 → 1000
 - 12 → 1100
 - 15 → 1111

Assignment 2.* (hint : use while)

- Print first 10 numbers
- Print the sum of first N numbers
 - N is provided by the user
- Find the sum of first N odd numbers
- Find smallest number divisible by all numbers between 1 to 20.
- Find the greatest common factor of 2 numbers .
 - 2,4 → 2
 - 6,9 → 3
- EASYMEDIUMHARD

Decision Making at a glance

- Branching
 - If/else
 - slides 30 - 32
 - switch
 - goto
- Looping
 - while
 - slide 45
 - do/while
 - for
- Jumps in loops
 - break
 - continue

Do-while loop

- Do-while is an exit controlled loop
 - (*while is entry controlled*)
 - When loop content should execute at least once
- E.g.
 - Looping until user enters valid value
 - Gather game input for game loop

```
do
{
    //loop body
} while (condition);
```

Do while : example

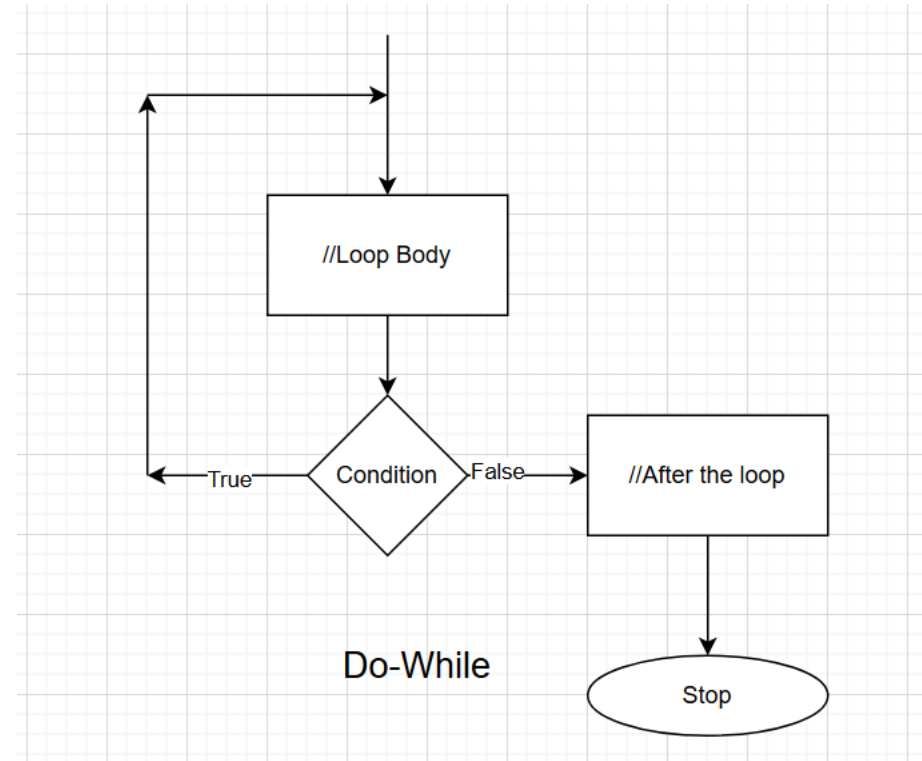
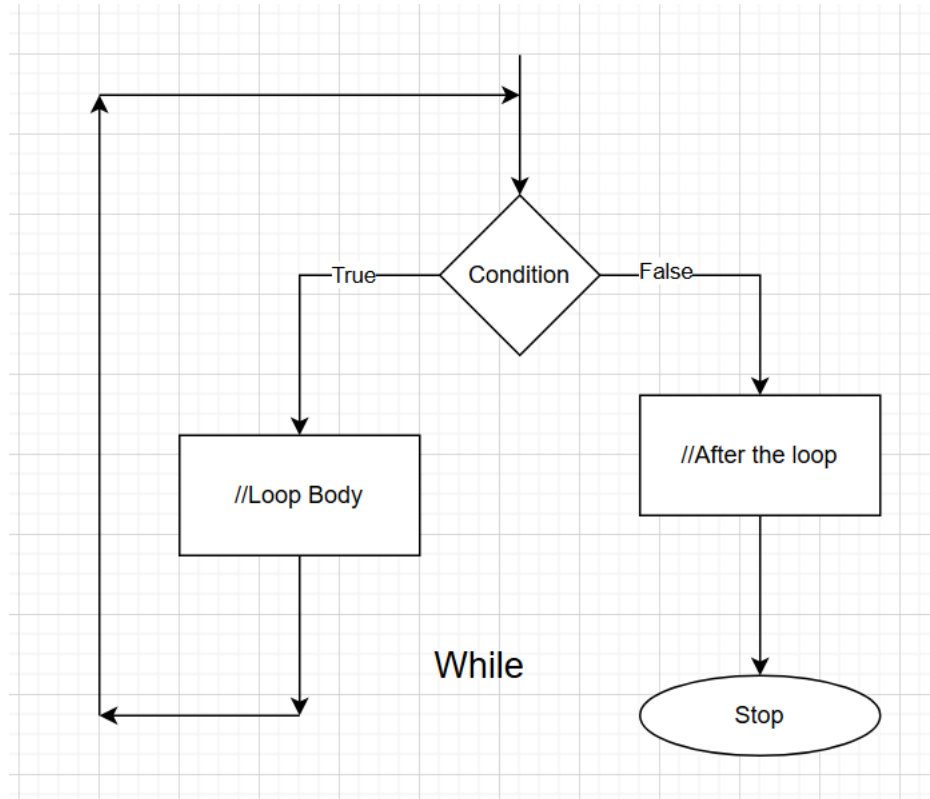
- Get a number between 1 and 4 from user.
 - Ask again if the numbers is not in range
 - Print the number and stop if it is in range.

```
int num = 0;

do
{
    cout << "\n Enter a number between 1 and 4 :";
    cin >> num;
} while ( num<1 || num >4);

cout << "\n You entered : " << num;
```

While vs do-while, visualized



Jumps in loops

- Break

- Jump to after the end of loop
- Loop does not execute

```
int num;
while (true) { // An infinite loop
    cout << "Enter a number (0 to stop): ";
    cin >> num;
    if (num == 0) {
        break; // Immediately exit the loop when the condition is met
    }
    //Process loop here
}
```

- Continue

- Jump to the beginning of the loop
- Loop continues to execute

```
do
{
    cout << "\n this is even number detector";
    cin >> num;

    if (num % 2 != 0)
    {
        cout << "\n this is not an even number";
        continue; //Continue executing the loop
    }
    cout << "\n this is an even number";
} while (true);
```

Assignment 1:

Create a **menu driven calculator**

- Enter 2 numbers
- Get user option (0 to 5) repeatedly
 - 1 → add the two numbers and print
 - 2 → subtract 2nd from first number and print
 - 3 → multiply the two numbers and print
 - 4 → divide 2nd by first number and print
 - **5 → do nothing , and restart**
 - 0 → stop the loop
- Try using both loops and compare result
 - while
 - Do/while
- Use **break** and **continue**

Boss Assignment:

Menu driven multi purpose program

- Create a multi purpose program running until user enters 0
- Enter 2 numbers e.g. a, b;
- Get user options repeatedly and perform the operation
 - 1 → find the least common multiple of 2 numbers
 - 2 → find the highest common factor of the 2 numbers
 - 3 → find 'a' raised to the power of 'b' using loops : a^b
 - 4 → find $1/a^b$ using loops
 - 5 → do something random from between 1 and 4
 - `#include<cstdlib>`
 - `int rand = 1+ rand()%4;`
 - 0 → Exit the program

Sequences

- Game leaderboard
- Scores per test
- Temperature per hour of day
- ...

Sequence of data types : Arrays

- `int days[7];`
 - Variables 'days'
 - Memory locations : $7 \times \text{sizeof}(\text{int}) = ?$
 - 1st : `day[0];`
 - 7th : `day [?]`
- `float avg[12]`
 - Variable 'avg'
 - Memory locations : $12 \times \text{sizeof}(\text{int}) = ?$
 - 1st : ?
 - 12th : ?

Array syntax

- `int days[7];`
 - one dimensional array
 - `[ ]` square brackets require a constant value
 - Index starts from 0
 - Access nth value using `days[n];`
 - Initialization
 - Option 1
 - `int days[7] = {1,2,3,4,5,6,7};`
 - Option 2 : omit the size
 - `int days[ ] = {1,2,3,4,5,6,7};`
 - Or GIGO ??
- * Array size is constant, cant change after creation.

Memory layout

- `int days[7] = {1,2,3,4,5,6,7};`

0	1	2	3	4	5	6
1	2	3	4	5	6	7

- Layout is Sequential
 - Next number is right after the previous ... all the way to end
- **Caution**
 - Do not go beyond the bound of array ... that is GIGO

Lab

- Q1:average of 5 numbers
 - Get 5 numbers from user
 - Print the average of the 5 numbers
 - Hints
 - `cin>>num[i];`
 - `sum = sum + num[i];`
 - loops
 - * Consider the right data type for average...
- Q2 : search whether a number exists in above array
 - (After printing average In the above entered numbers)
 - Get user to enter a search num
 - Return at what index location it exists in the array.

Names are sequence of characters

- `char name[100];`
- Characters arrays are '**null**' terminated
 - E.g `name = "ChatGPT";`
 - `if (name[7]== '\0')` would be true.
- So if names are array of chars...
 - What is array of names ??

Double dimension arrays

- `int nums[5][3] ;`
 - 5 row of 3 integers each
 - $5 \times 3 = 15$ total integers
- `char names[15][100];`
 - 15 names ... max size of 100 each.
 - $15 \times 100 = 1500$ characters
- How to create 10 rows of 100 floats each ?
 - What is the total number of floats used ?
 - What is the total memory size.

Memory layout : double dimension

- `int nums[5][3] ;`
 - 2 dimension = 2d Matrix
 - Rows are horizontal
 - Columns are vertical
 - Total $5 \times 3 = 15$ cells

	0	1	2
0			
1			
2			
3			
4			

Assignment 1

- Get 5 numbers from user
 - Print the highest number from the five.
 - E.g. user enters 5,6,2,3,7
 - Print
 - The highest number is : 7

Boss :Assignment

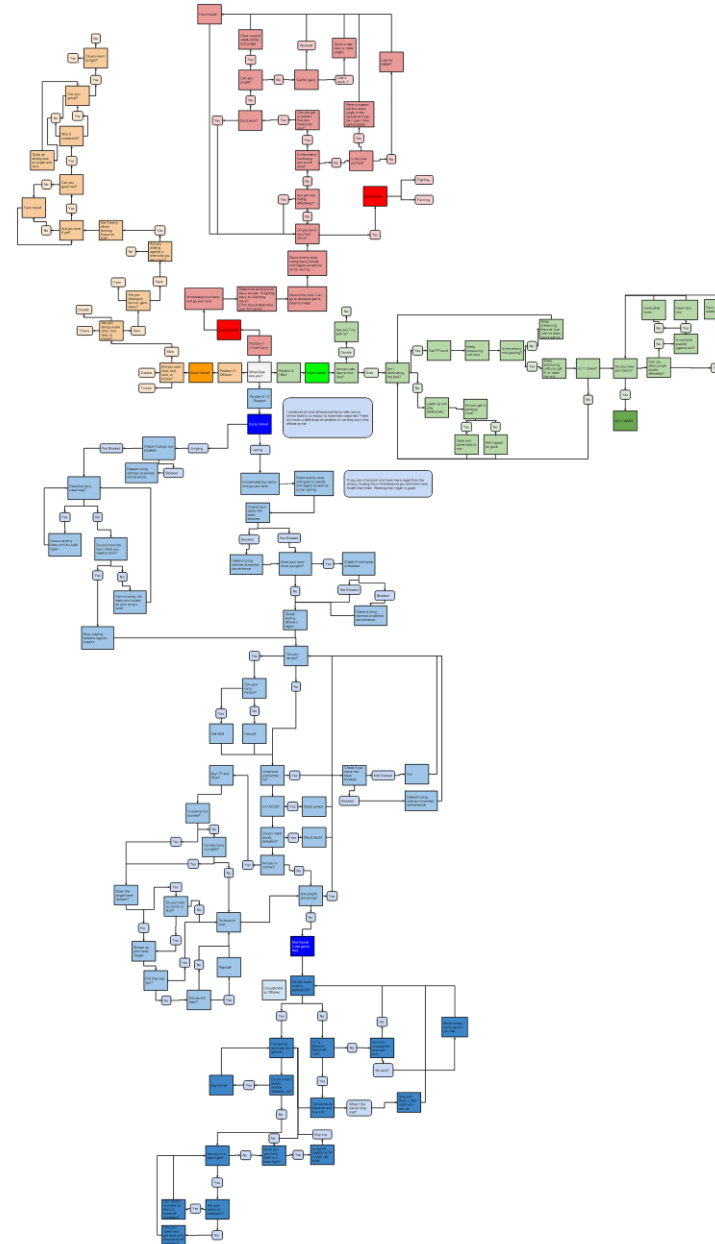
- Get 5 numbers from user
 - Print all the numbers in sorted order
 - E.g. user enters 5,6,2,3,7
 - Print
 - 2,3,5,6,7
- Hint
 - * would need multiple loops
 - * may need to know how to SORT data
 - *Sorts have not been taught ,do the best you can.*

Reuse, organize, simplify code

- How would you ...
 - Reuse the functionality that you have already implemented ?
 - Simplify your code that you can work on it independently
 - Organize your code for easy readability / maintainability ...
- Best programming Tool
 - Functions
- Best Principle
 - DRY → **D**on't **R**epeat **Y**ourself (i.e. in code 😊)

Functions : Need

- How to do you make spaghetti ?
 - Write a big enough program
- Who/what is most popular in C/C++
 - The main function
 - Nothing happens without the main ...
- Principle
 - Make spaghetti to eat but not in code.
 - Write a function , and it will be popular.



Functions : How ?

- Write a function
 - **Once**
- Use a function
 - **Many** times

```
int add(int x, int y)
{
    int sum = x + y;

    return sum;
}
int main()
{
    int num1 = 100;
    int num2 = 300;

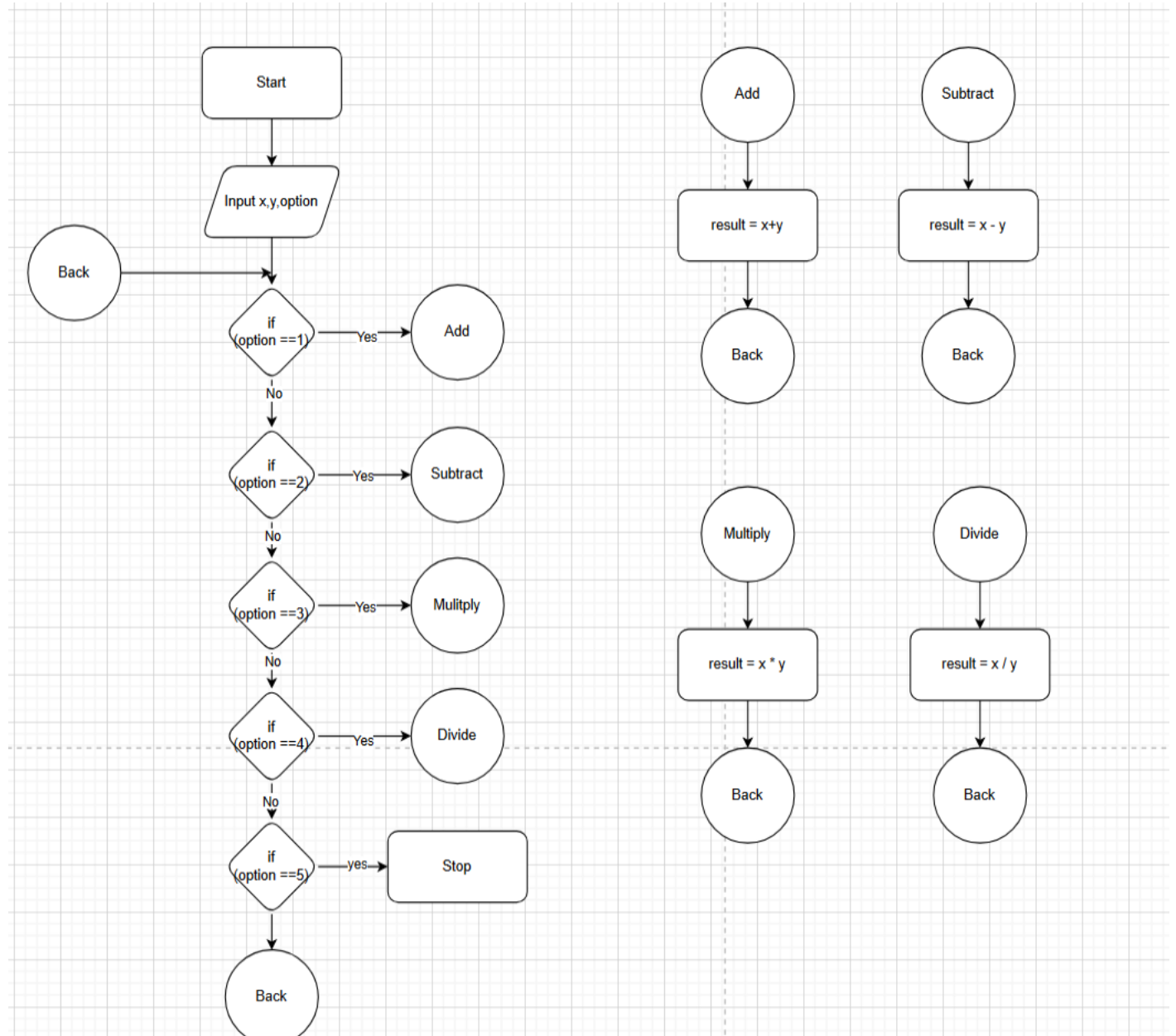
    int sum = add(num1, num2);
    int sum2 = add(1, 2);
    int sum3 = add(50, 100);
}
```

Lab : lets revisit calculator

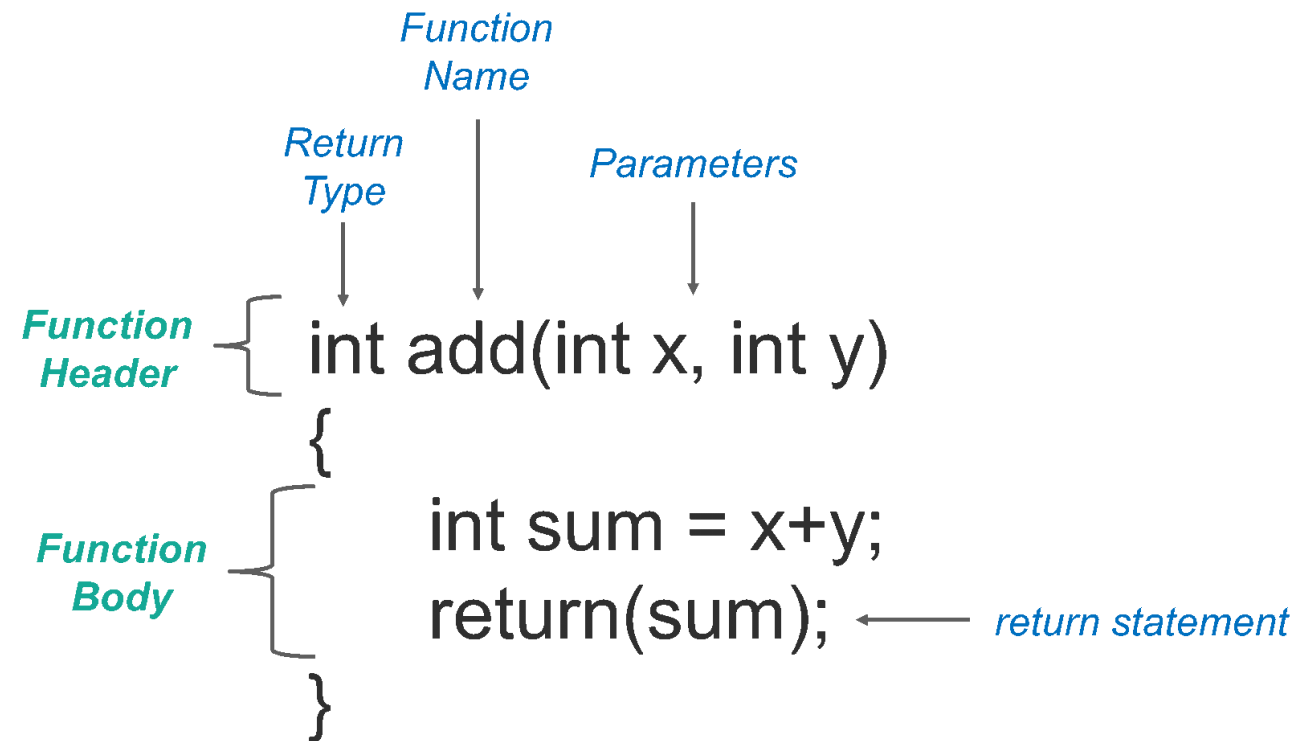
- Create a calculator
 - Take 2 numbers from user (x, y)
 - Take options 1 to 4 from user
 - Based on option print the following (**use functions**)
 - $1 \rightarrow x+y;$
 - $2 \rightarrow x-y;$
 - $3 \rightarrow x*y;$
 - $4 \rightarrow x/y$
 - $5 \rightarrow$ exit the program
 - $(\text{option} < 1 \ || \ \text{option} > 5) \rightarrow$ “chose between 1 to 4”

Flowchart View

- Function ~ Circle
- Return ~ Back Circle
- Lets code this ...



Anatomy of a function



More function nuances

- Function don't have to return a value
 - Then must use “void” as return type.
 - `void Print(int x) {..}`
- Function can exist without parameters
 - `void Print(){..}`
- Functions can call other functions (*nested functions*)
 - Like main calls add
- A function can call itself (*recursion*)

Assignment

- Write a program to convert temperatures
 - Celsius to Fahrenheit
 - and Fahrenheit to Celsius
- Use enters temperature and option
 - Option 1
 - Assume Celsius, Convert to Fahrenheit
 - e.g. 0 → 32
 - Option 2
 - Assume Fahrenheit, Convert to Celsius
 - e.g. 180 → 100
- Formula
 - $F = C * (180/100) + 32$
 - $C = (F - 32) * 100/180$
- **Must use functions**

Boss Assignment

- Fibonacci series : 0,1,1,2,3,5,8,13,21,34,55....
 - Starts with 0,1 and each number after is sum of previous 2 numbers
 - $0 + 1 \rightarrow 1$
 - $1 + 1 \rightarrow 2$
 - $8 + 13 \rightarrow 21$
- Task 1
 - Print n^{th} number from Fibonacci series using functions
- Task 2
 - Print each Fibonacci number all the way to n^{th} number
- Hints
 - Use functions
 - Use *appropriate* function parameters
 - Use recursive functions

string data type

- We already know few built in types
 - int , float , char, bool
- string
 - Is a derived data type,
 - Useful for names, description etc.
- Header
 - `#include<string>`
- Usage
 - `string name;`
 - `cin >> name;`
 - `cout <<name;`